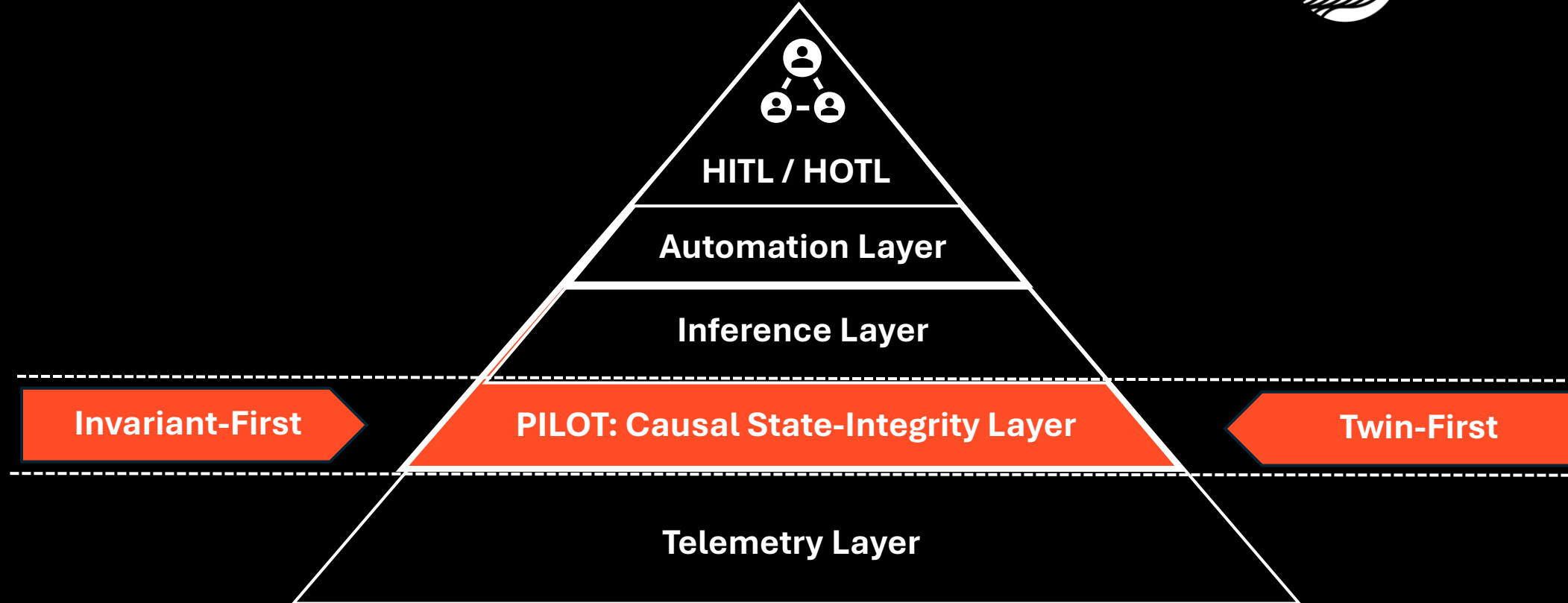


PILOT: The Solution



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PILOT is an out-of-band, deterministic solution. It detects inadmissible states which implies malicious activity, system fault, or data / model inconsistency. It validates the data that the higher layers operate on.

PILOT: Process Flow



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OT Telemetry

IT Telemetry

Ingestion:
Imports Data
System-State
Telemetry

STEP:
State Transition Extraction Pipeline
Turns messy telemetry into canonical
state transitions for validation

Deterministic Validation:
Exports Admissibility
Determines violations,
validations, ambiguity

**PILOT co-exists with and improves
existing cyber-security tools.
THERE IS NO RIP & REPLACE.**

Data Dependencies



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Data dependencies are determined by the control domain

IT Control Domains

1. Identity & authentication
2. Privileged access & admin
3. Endpoint execution
4. Cloud control plane
5. App deployment & CI / CD
6. Config & change management
7. Data access & movement
8. Automation & orchestration
9. Inter-domain trust & federation

Data Dependencies:

- **Events (Inputs)**
- **Identities (Authority)**
- **Mapping (Structure & Relationships)**
- **SOR (Grounding)**

OT SOR = Historian, PLC / DCS / SCADA
IT SOR = ID, Control-Plane, Config, Activity

OT Control Domains

1. Process physics
2. Control loop & automation
3. Actuator authority & command
4. Sensor integrity
5. Safety & interlock
6. Physical asset config
7. Op sequencing & workflow
8. Cross-process coupling
9. OT-IT interface & supervisory control

STEP



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STEP = State Transition Extraction Pipeline

Telemetry Normalisation

Critical Activities:

- Ingest heterogeneous telemetry from SORs
- Convert schemas to canonical format
- Mutation relevance & eligibility check
 - Standardise IDs, entities, time & event
- Temporal Alignment

State-Mutation Detection

Critical Activities:

- Normalised data v Authoritative State
- Determine if event changes the system
- If “**YES**”, then capture pre and post state
- Classify mutation
- If “**No**” then **IGNORE**

Canonical Transitions

Critical Activities:

- Bind Cause & Effect
- Encode the change
- Temporal Bounds
- Validate Consistency

Key Outputs:

- Entity_ID
- Transition_Type
- Pre_State
- Post_State
- Timestamp

Deterministic Validation



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